



MBotCube PVC 3D printer

able. 0.2mm and 0.3mm are the commonly available resolutions.

Horizontal resolution/XY resolution/feature resolution is the minimum displacement that can be made by a print head in the XY plane. It is measured in micrometres (μm) or dots per inch (dpi). Smaller feature resolution corresponds to finer features in the print.

High resolution or reduced error is particularly important if the casting is printed in parts and assembled separately. Low dimensional accuracy models may lead to misfit of parts whilst they are combined to create the final model.

Printing methods. There are several methods of 3D printing. These can be categorised into six core processes:

Photocuring.

Uses: Concept models, patterns and prototypes

For example: Stereolithography (SLA/SL), digital light processing (DLP), multi-jet modelling (MJM) and polyjet technology

Sintering.

Uses: Functional prototypes, low-volume production of complex components and investment casting

For example: Selective laser sintering (SLS) and direct metal laser sintering (DMLS)

Melting.

Uses: Functional prototypes, low-volume production of complex components and tooling components

Electron beam melting (EBM) and direct metal printing (DMP)

Lamination.

Uses: Wood-like models and concept models

For example: Laminated object manufacturing (LOM)

Extrusion.

Uses: Functional and durable models, production parts and manufacturing tools

For example: Fused dep-

ositive technology, more effective where multiple parts are required. It also offers increased accuracy which is needed for precision engineering applications, and showcases better print quality than high-end FDM machines.

Colour options. Extruder is the print head or the end of the printer where filament is melted and ejected. In extrusion method of printing, the number of print heads usually determines how many colours can be printed at once.

3D printers can be divided into three based on the colour options:

1. **Colour-choice printer.** These printers have only single extruder and can print objects only in one colour at a time. This colour is determined by the colour of the material used for printing, i.e., the object would be green in colour if green filament is used and red in colour if red filament is used.

2. **Basic-colour printer.** These models will have more than one extruder and can print a few colours in a single model.

3. **Full-spectrum colour printer.** Allows to print an object in a zillion different colours. This is achieved by mixing CMYK colour model—cyan, magenta, yellow and key (black) and jetting it on each thin, plaster-based layer.

Filaments of all primary and secondary colours are easily obtained in the market. In addition to these, exotic colours such as gold, silver and fluorescent are also available.

Cost of parts. Cost of parts is expressed in cost per unit volume. Cost of parts is provided by vendors for a specific part, or an average across a group of different parts. These need not necessarily represent the true cost of printing as they exclude the cost of support materials, process waste generated and other consumables used in



Stratasys Objet1000 3D printer

osition modelling (FDM)/fused filament fabrication (FFF), plastic jet printing (PJP) and thermoplastic extrusion jetting.

Uses: Large objects, fast and cheap models and smooth models with lots of fine details

For example: Powder bed and inkjet head/3D printing (3dp), MJM, smooth curvature printing (SCP) and DMP

The key advantage of FDM is reusability. Since raw thermoplastic materials are used here, this technology is less susceptible to dimensional changes, unlike other technologies. This technology offers easy material changeover. FDM has fair levels of accuracy which fulfils the needs of a basic perspective making it the most popular in low-end, economical devices. But this method mostly lacks the resolution and finish of a professional-grade output.

SLS is a very expensive but much